

Model	Actual Text	Speed	.wav files	.m4a files
Tiny	හැබැයි කඩය යට වෙලා තියෙනවා.අපි ඒගොල්ලන්ගේ building එක හදන්න දෙන්න ඕනි	5.08 seconds	❖rrr e .	HessenO êgel Netz 物 と y rooms Fahr 工康 reasons 模仿 Corporation 及 Mov 平 形 物圖源 湊珍 物 siècle tre kr 忝 保 休 生 物 平 形 根 明 形 自 生 Steam 針 一 形 另 平 形sted 一 形 一 現 прот 生 石 ツ 彌 形 一 其 原 原 明 現 永 系 一 真 落 深 Reports 分 ん erek 秀 平 形 一 形 一 現 針 一 形 一 形 一 形 一 形 一 Zig 伊 土 青 色 oma практически 形 一 土 áe 針 土 罽 仍 明 簽 zem 學 墨 形 一 形 一 蜒 蘆
Base		16.82 seconds	...rota we make di na, heba'i ay kadeayata vela da Ginauxu, ...hapi ei ag 迎 ala ng built inna ke haat ni dineknu da dinebuna!	ndha hai kaadayatavilaatinu. ndha hai kaadayatavilaatinu. ndha hai kaadayatavilaatinu.
Small		51.04 seconds	ව කාපි මෙත❖甚බරන් ො ව ග තින් න❖ Spielන ත□ pen	හොඳ්ක්ශරය එකෙමතු □ලන්භ ඉ MA、 කර ඇතන්තිට්මාau න❖ speeds plus ❖orde හසලසසන්රරයගේ
Tiny	මම ගෙදර යනවා	2.45 s	hh	M实k
Base		3.40 s	マム・ゲット ヤ kunne made	实物作り
Small		4.12 s	Mame, gjennom en yan	Mamme, gjennere, gjennemmer!
Tiny	My name is Linda Parker. I am 59 years old.	22.88 seconds(m4a)	30.29 seconds	My name is Linda Parker. I am 59 years old.
Base		35.86 seconds(m4a)	38.38 seconds	
Small		98.21 seconds(m4a)	101.06 seconds	

Whisper Model	File Type	Accuracy / Output Quality	Processing Speed	Observations
Tiny	.wav	Very poor for Sinhala. Output contained random symbols and unrelated characters.	Fastest	Cannot properly recognize Sinhala speech.
Tiny	.m4a	Extremely poor recognition. Mostly garbage text/symbols.	Fast	Low accuracy for compressed audio.
Base	.wav	Slightly better than Tiny. Some Sinhala sounds recognized phonetically but not accurate text.	Moderate	Partial speech understanding but unreliable.
Base	.m4a	Poor recognition with repeated incorrect phrases.	Faster than .wav in some cases	Compression affects recognition quality.
Small	.wav	Better English recognition. Sinhala still inaccurate but more understandable than Tiny/Base.	Slow	Higher accuracy but large processing time.
Small	.m4a	Slightly improved compared to Tiny/Base but still weak for Sinhala.	Slow	Better language understanding but compressed audio reduces performance.

Conclusion

Aspect	.wav Files	.m4a Files
Audio Quality	Higher quality	Compressed quality
Sinhala Recognition	Better than .m4a	Poorer performance
Noise Handling	More stable	More distorted outputs
Processing Speed	Slightly slower	Sometimes faster
Accuracy	Higher transcription accuracy	Lower transcription accuracy

File Name	Format	Sampling Rate (Hz)	Bitrate	Channels	Duration (s)	File Size (MB)
audio1.m4a	.m4a	16000	256000	1	5	0.15
audio1.wav	.wav	16000	256000	1	5	0.15
audio2.m4a	.m4a	48000	0	0	5.29	0.06
audio2.wav	.wav	48000	0	0	5.29	0.06
audio3.m4a	.m4a	16000	256000	1	5	0.15
audio3.wav	.wav	16000	256000	1	5	0.15
audio4.m4a	.m4a	16000	256000	1	5	0.15
audio4.wav	.wav	16000	256000	1	5	0.15
audio5.m4a	.m4a	44100	48044	2	14.95	0.09
audio5.wav	.wav	44100	48044	2	14.95	0.09
audio6.m4a	.m4a	48000	0	0	2.94	0.04
audio6.wav	.wav	48000	0	0	2.94	0.04
audio7.m4a	.m4a	48000	0	0	3.95	0.05
audio7.wav	.wav	48000	0	0	3.95	0.05
audio8.m4a	.m4a	44100	0	2	482.79	2.81
audio8.wav	.wav	44100	0	2	482.79	2.81

Observation

- Multiple sampling rates were identified in the dataset. (16 kHz,44.1 kHz,48 kHz)
- Whisper is optimized for 16 kHz mono audio.
- Mono audio files were mainly recorded at 16000 Hz.Stereo audio appeared in higher sampling rate files.Some compressed files did not properly expose metadata.
- Some .m4a and .wav files had unavailable bitrate information.
- Most paired .wav and .m4a files had similar technical properties.